



ANALYSTLAB AFRICA

Data Science Internship Programme

Official Weekly Assignment

Week 3: Advanced Data Exploration, Statistical Analysis & Feature Engineering

Document Control

Programme	Data Science Internship
Week	Week 3
Project	Advanced Data Exploration, Statistical Analysis & Feature Engineering
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 3 (Sunday, 11:59 PM WAT)

Business Scenario

Following your data cleaning and exploratory data analysis in Week 2, your organization has approved your dataset for deeper statistical investigation before predictive modelling begins.

As a Junior Data Scientist, your responsibility is to uncover hidden patterns, validate business assumptions using statistical analysis, engineer meaningful features, and prepare a machine learning-ready dataset.

Your findings will provide business stakeholders with a deeper understanding of the factors influencing the target variable while producing a high-quality dataset ready for machine learning development in Week 4.

Learning Objectives

By the end of this assignment, you should be able to:

- Perform advanced exploratory data analysis (EDA).
- Apply statistical techniques to validate hypotheses.
- Engineer meaningful features for predictive modelling.
- Identify relationships between variables.
- Select relevant features for machine learning.
- Prepare a modelling-ready dataset.
- Translate statistical findings into business insights.

Project Options

Continue working on the **same dataset selected in Week 2**.

Option 1 – Heart Disease Prediction Dataset

Dataset:

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Objective

Prepare and analyze the dataset to identify the key factors associated with heart disease and optimize the data for predictive modelling.

Option 2 – Loan Prediction Dataset

Dataset:

<https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset>

Objective

Prepare and analyze the dataset to identify the major factors influencing loan approval decisions and optimize the data for predictive modelling.

Assignment Tasks

Part 1 – Advanced Data Quality Assessment

Review your Week 2 cleaned dataset.

Evaluate:

- Missing values
- Duplicate records
- Data consistency
- Outlier detection
- Feature distributions
- Class imbalance
- Data quality improvements made

Document all findings.

Part 2 – Advanced Exploratory Data Analysis (EDA)

Perform a comprehensive exploratory analysis.

Include:

Numerical Analysis

- Histograms
- Boxplots
- Density plots
- Distribution analysis

Categorical Analysis

- Frequency distributions
- Cross-tabulations
- Percentage analysis

Bivariate Analysis

Analyze relationships between:

- Numerical vs Numerical
- Numerical vs Categorical
- Categorical vs Categorical

Multivariate Analysis

Investigate complex relationships among multiple variables using appropriate visualization techniques.

Part 3 – Statistical Analysis

Perform at least **four statistical analyses** appropriate for your dataset.

Examples include:

- Correlation Analysis
- Chi-Square Test
- Independent T-Test
- ANOVA
- Mann-Whitney U Test
- Kruskal-Wallis Test

For each statistical test, include:

- Objective
- Null hypothesis (H_0)
- Alternative hypothesis (H_1)
- Test statistic
- P-value
- Interpretation
- Business implication

Part 4 – Feature Engineering

Create new features that could improve predictive performance.

Examples include:

Heart Disease

- Age Groups
- BMI Categories (if applicable)
- Cholesterol Risk Level
- Blood Pressure Category
- Combined Risk Score

Loan Prediction

- Income Bands
- Debt-to-Income Ratio
- Family Size Groups
- Employment Categories
- Credit Risk Category

Explain:

- Why each feature was created
- Expected business value
- Expected modelling benefit

Part 5 – Feature Selection

Evaluate every feature in the dataset.

Identify:

- Important features
- Highly correlated variables
- Redundant features
- Features to remove

Support your decisions using statistical evidence.

Part 6 – Data Preparation for Machine Learning

Prepare the final modelling dataset.

Include:

- Encoding categorical variables
- Feature scaling (where applicable)
- Final cleaned dataset
- Machine learning-ready dataset
- Updated data dictionary

Save the processed dataset.

Part 7 – Business Insights Report

Prepare a **3–5 page report** summarizing:

- Key statistical findings
- Significant relationships
- Feature engineering decisions
- Important predictors
- Business recommendations
- Dataset limitations
- Next steps for predictive modelling

Minimum Project Requirements

Exploratory Data Analysis

Your notebook **must** include:

- At least **20 visualizations**
- Distribution analysis
- Correlation heatmap
- Pair plots or equivalent multivariate visualization
- Outlier analysis
- Class distribution analysis

Statistical Analysis

Your project **must** include:

- At least **4 statistical tests**
- Clearly stated hypotheses
- Test statistics
- P-values
- Business interpretation

Feature Engineering

Create at least **5 engineered features**.

Explain:

- Why they were created
- Expected modelling benefit

Feature Selection

Include:

- Correlation analysis
- Selected features
- Removed features
- Justification

Documentation

Your notebook should contain:

- Clear markdown explanations
- Well-commented code
- Professional formatting
- Business interpretations
- Tables where appropriate

GitHub Repository

Your repository should include:

- README.md
- Jupyter Notebook
- Business Insights Report
- Statistical Analysis Report
- Data Dictionary
- Final Clean Dataset
- Requirements.txt

Deliverables

Submit the following:

1. Jupyter Notebook (.ipynb)
2. Business Insights Report (PDF or Word)
3. Statistical Analysis Report (PDF or Word)
4. Feature Engineering Documentation
5. Updated Data Dictionary
6. Final Machine Learning Dataset (.csv)
7. README.md
8. GitHub Repository
9. Google Drive Folder (Anyone with the link can view)
10. Submit using the official Week 3 Submission Form

Assessment Criteria

Criteria	Weight
Business Understanding & Statistical Thinking	15%
Exploratory Data Analysis	25%
Statistical Analysis	20%
Feature Engineering & Feature Selection	20%
Business Insights & Interpretation	10%
Documentation & Professionalism	10%

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 3 project.
- Publish a LinkedIn post describing your advanced data exploration, statistical analysis, and feature engineering process.
- Share the most important insights from your analysis and explain how they prepare the dataset for predictive modelling.
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafri**c on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

Professional Data Scientists do more than clean data, they uncover patterns, validate assumptions, engineer meaningful features, and prepare reliable datasets that drive predictive models. This assignment is designed to simulate real-world data science work, emphasizing analytical reasoning, statistical thinking, and business communication. Present your findings as though you are reporting to healthcare professionals or financial decision-makers, depending on your chosen project.

Submission Checklist

- ☐ Jupyter Notebook (.ipynb)
- ☐ Business Insights Report
- ☐ Statistical Analysis Report
- ☐ Feature Engineering Documentation
- ☐ Updated Data Dictionary
- ☐ Final Machine Learning Dataset (.csv)
- ☐ README.md Updated
- ☐ GitHub Repository Updated

☐ LinkedIn Post Published

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